

Module 02: Agents -- You as a Prediction Machine

Learning Objectives

1. Define an **agent** in Active Inference as any system that maintains a generative model and acts to minimize free energy.
2. Explain how your sense of **personal identity** functions as a generative model -- a set of predictions about who you are and how you will behave.
3. Analyze the relationship between **autonomy**, **agency**, and the capacity to select actions based on internal models.
4. Reflect on how adolescence involves a fundamental reconstruction of the self-model.

Introduction

What makes you *you*? Not your atoms -- those are replaced constantly. Not your memories -- those are reconstructed every time you recall them. Not your personality -- that shifts depending on context. Yet something persists. You have a continuous sense of being a specific person with a particular history, specific preferences, and a trajectory into the future.

Active Inference offers a striking answer to this puzzle: *you* are a generative model. More precisely, you are the process of a particular generative model maintaining itself over time -- predicting, acting, and updating in ways that preserve a coherent pattern of engagement with the world. Your sense of self is not a thing but an ongoing process of inference.

Key Concepts

1. What Is an Agent?

In Active Inference, an agent is any system that has a **Markov Blanket** (a boundary separating it from its environment) and a **generative model** (internal states that encode predictions about sensory inputs and the consequences of actions). Crucially, an agent does not just passively model the world -- it *acts* on the world to make its predictions come true. You do not just predict that you will eat lunch; you act to make lunch happen.

2. The Self as a Generative Model

Your identity -- your sense of who you are -- is a generative model about yourself. It includes predictions like "I am good at math," "I am shy around strangers," "I care about fairness." These predictions shape what you notice, how you interpret events, and what actions you take. When evidence contradicts your self-model (you fail a math test, you speak confidently to a stranger), you experience prediction error that must be resolved -- either by updating your self-model or by dismissing the evidence.

3. Agency and Autonomy

Agency is not all-or-nothing. Active Inference defines agency in terms of the degree to which a system's actions are driven by its own internal models rather than by external forces. A puppet has no agency; its movements are entirely determined from outside. A student who chooses classes based on genuine curiosity has high agency. A student who chooses classes solely based on what friends are taking has less. Agency is the extent to which your generative model, rather than external pressures, drives your behavior.

4. The Adolescent Agent in Transition

Adolescence is a period of intense agency development. Your generative model is being revised in fundamental ways: who am I, what do I value, what kind of person do I want to become? This process is uncomfortable precisely because it involves high prediction error -- the old model of yourself (the child you were) no longer fits, and the new model (the adult you are becoming) is not yet stable. Active Inference predicts exactly this discomfort: model revision is cognitively expensive.

5. Multiple Selves, One Agent

You probably act differently around your parents, your friends, and your teachers. Does this mean you have multiple identities? Active Inference would say you have one generative model with **context-dependent predictions**. Your model includes predictions about how to behave in different settings. This is not hypocrisy -- it is sophisticated inference about what different social environments expect. The unity of the agent lies not in behaving identically everywhere but in maintaining a coherent model that generates appropriate predictions across contexts.

Active Inference Connection

The free energy principle implies that agents must continuously minimize the discrepancy between their predictions and their sensory evidence. For the self-model specifically, this means you are always working to maintain a coherent identity. When your self-model predicts "I am a kind person" and you catch yourself being cruel, the prediction error generates discomfort (what psychologists call **cognitive dissonance**). You resolve it either by updating your model ("maybe I am not always kind") or by reinterpreting the evidence ("they deserved it"). Both strategies minimize free energy; only the first involves genuine model updating.

Applications

- **Case Study 1 -- The College Application Essay:** Writing a personal statement forces you to articulate your generative self-model explicitly. Why is this so difficult? Because your self-model operates largely below conscious awareness -- it generates predictions automatically. Putting it into words requires making the implicit explicit, which is itself a form of model inspection. The students who struggle most are often those whose self-model is in the most active revision.
- **Case Study 2 -- Identity and Social Media Personas:** Your social media profile is an externalized generative model -- a public set of predictions about who you are. But the feedback loop between your online persona and your self-model is powerful. Likes and comments become sensory evidence that either confirms or disconfirms your self-model. This is why social media can be both identity-affirming and identity-destabilizing: it dramatically increases the rate of prediction error feedback about who you are.

Discussion Questions

1. Think of a time when you discovered something about yourself that surprised you. What prediction did your self-model get wrong, and how did you update it?
2. How much of your sense of identity is truly your own generative model versus a model constructed by the expectations of people around you? Where do you draw the line between internal and external influence?

3. If identity is a generative model rather than a fixed essence, does that make it more or less "real"?

Summary

An agent, in Active Inference, is a system that maintains a generative model and acts to minimize free energy. Your sense of self -- your identity -- is the most personal instance of this: a continuously updated model of who you are, what you value, and how you will behave. Adolescence is a period of intensive self-model revision, which is why it feels both exciting and unsettling. In Module 03, we turn to **perception** -- how your generative model interprets the flood of sensory information from your social world.